

Project ID:

25-26J-323

1. Topic (12 words max)

An AI-based web application for Skin disease prediction & Suggests solutions.

2. The research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max)
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- references not included in word count.

Millions of people worldwide suffer from skin diseases, which range widely in severity from minor ailments like eczema to severe conditions like melanoma (Esteva et al., 2017). Although prompt diagnosis is crucial, access to dermatological care is still restricted in rural or underdeveloped areas. Furthermore, manual diagnosis can lead to misdiagnosis or delayed treatment because it is frequently subjective and can vary based on the clinician's level of expertise (Tschandl et al., 2020).

Automated skin image analysis has the potential to support dermatological diagnosis as artificial intelligence (AI) advances, particularly in the fields of machine learning (ML) and deep learning (DL) (Brinker et al., 2018). This research aims to develop a comprehensive system for predicting skin diseases.

The main component entails training models, especially convolutional neural networks (CNNs), to classify skin images into distinct disease categories, such as bacterial (like impetigo), viral (like herpes simplex), parasitic (like scabies), and fungal (like ringworm). CNNs' strong image recognition and classification performance make them ideal for these tasks (Esteva et al., 2017). This system will use federated learning to improve data security and privacy by enabling AI models to be trained locally on user devices without sending private images to a central server (Sheller et al., 2020). This approach allows the model to learn and improve across a user network while protecting personal health data.

Additionally, the system will provide individualized skincare and treatment recommendations. These could be lifestyle advice based on the user's diagnosis, dermatologist recommendations, or over-the-counter treatments. The ultimate objective is to develop an intelligent, private, and easily accessible AI-based platform that supports the early diagnosis and treatment of skin conditions; this will be especially helpful for underserved or remote communities.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

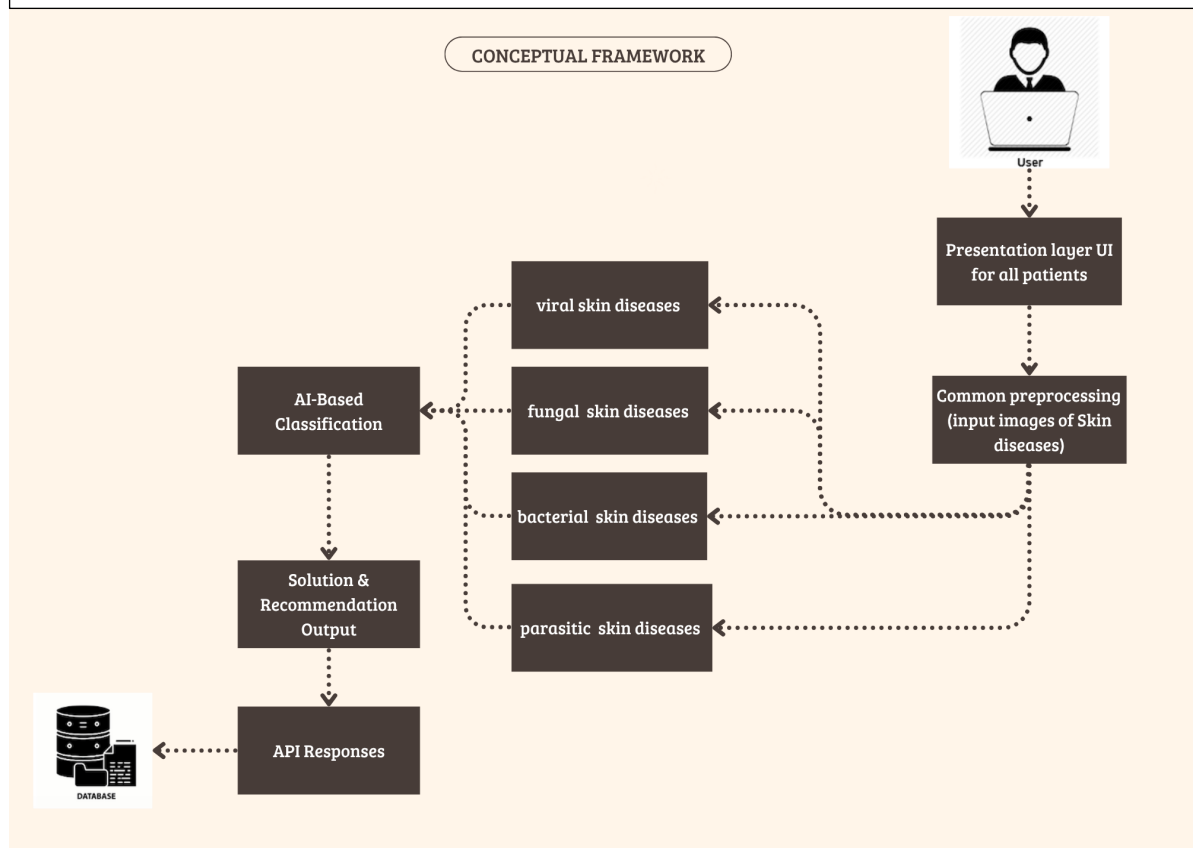
This study suggests an AI-powered, privacy-preserving method for identifying skin conditions and suggesting appropriate treatments. The solution analyzes skin images and provides precise, individualized health feedback by combining three key technologies: Federated Learning (FL), Deep Learning (DL), and Machine Learning (ML).

Using a mobile or web interface, the system first gathers skin images uploaded by users or taken by cameras. A trained Deep Learning model, like a Convolutional Neural Network (CNN), uses these preprocessed images (resizing, denoising, and feature extraction) to classify the skin condition (e.g., eczema, melanoma, impetigo, etc.).

Federated Learning trains the model across several user devices without sending raw data to a central server, improving privacy. Model updates, not photos, are exchanged and combined to increase accuracy while maintaining data security.

Once a disease has been predicted, the system offers tailored advice, such as skincare regimens, suggested treatments, and notifications for medical consultation. The system changes in response to user input and revised model weights as time passes.

With its accessible, affordable, and clever alternative for early skin disease detection and care, this solution is especially helpful for rural communities with limited dermatologist access.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Domain Expertise: The present research calls for multidisciplinary knowledge in artificial intelligence and dermatology. To correctly interpret visual symptoms and annotate datasets, dermatologists must thoroughly understand the different types of skin diseases, including bacterial, viral, fungal, parasitic, and autoimmune skin conditions. Skilled dermatologists are also essential for validating image labels, detecting uncommon skin conditions, and defining disease characteristics used in model training. Working with medical experts guarantees that the AI model supports real-world decision-making and complies with clinical diagnostic standards.

Specialized Knowledge: Machine learning (ML), deep learning (DL), and federated learning (FL) are key components of this research in terms of technical expertise. Proficiency in computational neural networks (CNNs) for image classification, model evaluation metrics (like precision, recall, and AUC), and overfitting mitigation techniques is essential. Additionally, knowledge of client-server coordination, privacy-preserving algorithms, edge AI deployment, and decentralized model training is necessary for FL implementation. To properly build and optimize the solution, one must also be proficient with tools like TensorFlow, PyTorch, Python, and OpenCV.

Data requirements: A sizable, varied, and annotated dataset of skin photos that represent various skin tones, lighting scenarios, and disease classifications is needed for the study. The baseline sources will be public datasets. However, with user consent, more data will be gathered via user devices to enhance generality and facilitate federated learning. To guarantee clinical accuracy, medical professionals must label every image. To support contextual diagnosis, the dataset should contain metadata such as patient age, affected body part, and symptom duration, if available.

8. Objectives and Novelty

Main Objective To develop a comprehensive, AI-powered web and mobile application that predicts various skin diseases using ML, DL, and FL, and provides secure, personalized healthcare solutions.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Vivushan T IT22117700	Design the image acquisition and prediction module for viral skin diseases and recommendation module for viral skin diseases	- Develop image capture and preprocessing using OpenCV - Apply ML and CNN-based models to detect viral diseases (e.g., herpes, measles) - Integrate Federated Learning for privacy-preserving training on mobile - Handle mobile UI for image submission and results	Combines real-time image processing, ML/DL classification, and on-device federated learning for viral skin detection in a seamless mobile interface
Afzal Ahamed A H IT22102782	Develop AI-based diagnosis system for fungal skin diseases and recommendation module for fungal skin diseases	- Preprocess skin images using augmentation and segmentation - Build CNN/ML models to detect diseases like ringworm or athlete's foot - Integrate FL to enhance model generalization across devices - Enable results visualization with backend-frontend sync	Adaptive DL system trained with federated updates across clients, tailored to recognize subtle fungal patterns in diverse environments
Buyravy P IT22080776	Implement secure prediction pipeline for bacterial skin diseases and recommendation module for bacterial skin diseases	- Create preprocessing layer for bacterial symptom images - Train ML/DL models for diseases like impetigo and cellulitis - Set up FL server-client architecture with real-time update logic - Manage dashboard for results tracking	Implements end-to-end bacterial skin diagnosis pipeline secured via federated learning and integrates result monitoring with visual feedback
Vinthiya P IT22080844	Build intelligent solution and recommendation module for parasitic skin diseases	- Use ML and DL to classify scabies, lice, etc. - Apply image preprocessing to enhance detection - Aggregate FL-trained model outputs for consistency - Generate personalized treatment suggestions and UI alerts	Recommendation engine linked to FL-trained outputs and user feedback loop to continuously update solution accuracy and care advice

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22117700	Focuses on viral skin diseases like herpes, chickenpox, and measles. Applies knowledge in mobile application development and computer vision to implement image capture, preprocessing, and user interface for viral disease detection. Aligns with the specialization by integrating front-end development with AI-based prediction modules.
IT22102782	Works on fungal skin diseases such as ringworm and athlete's foot. Trains and evaluates deep learning models using Convolutional Neural Networks (CNNs) to classify fungal infections. This aligns with machine learning and image processing specialization through the application of DL algorithms in medical imaging.
IT22080776	Handles bacterial skin diseases like impetigo and cellulitis. Designs and deploys federated learning (FL) models for privacy-preserving disease prediction. This complies with the specialization in artificial intelligence and distributed systems by applying FL techniques for secure model training.
IT22080844	Responsible for parasitic skin diseases such as scabies and lice. Develops the recommendation system that delivers treatment tips and alerts based on model output. Uses backend development and intelligent systems expertise, aligning with the software engineering and AI specialization areas.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Harinda	Sahadeva Fernando	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes ☒ No ☐

- b) Does the proposed topic exhibit novelty?

Yes ☒ No ☐

- c) Do you believe they have the capability to successfully execute the proposed project?

Yes ☒ No ☐

- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes ☒ No ☐

- e) Supervisor's Evaluation and Recommendation for the Research topic:

OK.

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

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Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.